MIL-S-51421(EA)
AMENDMENT 1
10 September 1986

MILITARY SPECIFICATION

SCREENING, CHEMICAL-BIOLOGICAL FILTER

This amendment forms a part of MIL-S-51421(EA), dated 8 March 1976, and is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Add in sequence:

"2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

ASTM STANDARDS

G 21 - Determining Resistance of Synthetic Polymeric Materials to Fungi

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)"

PAGE 4

4.1.2:

Renumber 4.1.2 to 4.1.3 and add in sequence new 4.1.2:

"4.1.2 Responsibility for mildew resistance test. The contracting officer shall select either the contractor's, the Government's, or the National Center for Toxicological Research's (NCTR) facility, as specified (see 6.2), to perform the mildew resistance test in accordance with 4.4.13."

MSC N/A FSC 4240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

PAGE 9

Table V. Test methods, at the line "Yarn slippage", under "Number determinations per sample unit": Delete "3" and substitute "5".

PAGE 10

Table V. Test methods, at the line "Mildew resistance", under "Results reported as": Delete "Growth or no growth" and substitute "Rating scale".

PAGE 11

4.4.10, line 6, after sentence 4 add: "The jaws of the tensile testing machine shall be centered on the specimen."

PAGE 13

4.4.13: Delete sentence 1 and substitute "Determination of the mildew resistance shall be in accordance with ASTM G 21, Determining Resistance of Synthetic Polymeric Materials to Fungi."

4.4.13(a): Delete in its entirety and substitute:

"(a) Maintaining stock cultures. Cultures of these fungi shall be maintained separately on a potato-dextrose agar medium. The stock culture may be kept for not more than four months in a refrigerator at approximately 3° to 10°C (37.4° to 50°F). Subcultures incubated at 23° to 30°C (82.4° to 86°F) for 7 to 20 days shall be used in preparing the inoculum. For extended storage of stock cultures, ATCC cultures may be suspended in broth and dispensed in 5-10 aliquots into screw-capped test tubes. The suspensions shall then be frozen at -50°C and held at that temperature until needed, i.e., one tube of each culture will be used every four months to initiate stock cultures for testing during that period."

4.4.13(b): Delete in its entirety and substitute:

"(b) Test specimens. Three (3) specimens, 2 inch square taken at random shall be cut from the sample, completely immersed in slowly running water for 24 hours, and subsequently dried. The leaching tank shall be large enough to hold an amount of water weighing not less than 50 times the weight of the specimens, the water shall not fall directly on the specimens, and shall flow at a rate of 5 to 10 liters per hour. The pH of the water shall be in the range of 6.0 to 8.0."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

4.4.13(c): Delete in its entirety and substitute:

"(c) Evaluation of results. Upon completion of the incubation period, the test specimens shall be examined visually. The treated screening material shall be rated in accordance with ASTM G 21, and shall be considered to have failed when any of specimens has a rating of 1 or greater."

4.4.13(d): Delete in its entirety.

PAGE 15

6.2: Add in sequence:

"(e) Mildew resistance (see 4.1.2)."

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A870

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.